

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
I SEMESTER 2025-26
CS/ECE/EEE/INSTR F241, Microprocessor & Interfacing
25-03-2025 (Wednesday) QUIZ-4(closed book)

A

Max time: 10 Min

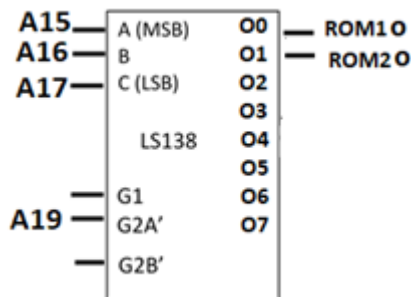
Max Marks: 08

ID No:

Name:

Sec No.:

Q1. An 8086 processor is interfaced with 64KB ROM memory (starting address from 40000H) through a decoder as shown below. 16K memory chips are available. This interfacing uses absolute addressing. Two 74LS138 are used for this interfacing (one for generating chip select for odd bank, the other for even bank), and no other gates are available. System bus control signals are used for memory interfacing. [1+2+2+1M]



Answer the following questions.

Question	Answer
What should be connected to G1 ?	A18
What should be connected to G2B' ?	BHE'
Mention the last address from 8086 that will be accessed in ROM2O (20 bit address in hex)	4FFFF
Which address lines are connected to the 16KB ROM1O Chip (mention the range)	A1-A14

Q2. For an 8086-based system, the following parameters are given:

- Address setup time = 40 ns , Data setup time = 100 ns , Clock period = 200 ns
- The memory device provides valid data **1100 ns** after the address becomes valid

Determine the number of wait states required. Marks will be awarded only if justification is correct [2M].

No of wait states	4
Justification	

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B

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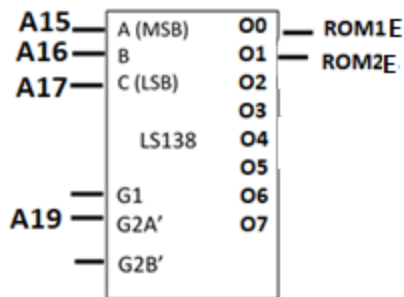
Max Marks: 08

ID No:

Name:

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Q1. An 8086 processor is interfaced with 64KB ROM memory (starting address from 40000H) through a decoder as shown below. 16K memory chips are available. This interfacing uses absolute addressing. Two 74LS138 are used for this interfacing (one for generating chip select for odd bank, the other for even bank), and no other gates are available. System bus control signals are used for memory interfacing. [1+2+2+1M]



Q1. Answer the following questions.

Question	Answer
What should be connected to G2B'?	A0
What should be connected to G1?	A18
Mention the last address from 8086 that will be accessed in ROM1E (20-bit address in hex)	47FFE
Which address lines are connected to the 16KB ROM1E Chip (mention the range)	A1-A14

[2M]

Q2. For an 8086-based system, the following parameters are given:

- Address setup time = 40 ns , Data setup time = 100 ns , Clock period = 200 ns
- The memory device provides valid data **1300 ns** after the address becomes valid

Determine the number of wait states required. Marks will be awarded only if the justification is correct [2M].

No of wait states	5
Justification	